Results: Field Overview

Results: Field Overview I

The de-duplicated corpus for **Exoplanet Atmospheres** contains $N = 46$ records spanning 2000–2023 (23 years). Publication volume grows at a compound annual rate of 6.76% (mean year-over-year growth 38.9%, doubling time 2.1 years), peaking in 2023 with 9 records that year. The growth curve is a first-order descriptive summary of the retained corpus; it is not a field-wide estimate of research activity.

Results: Field Overview I

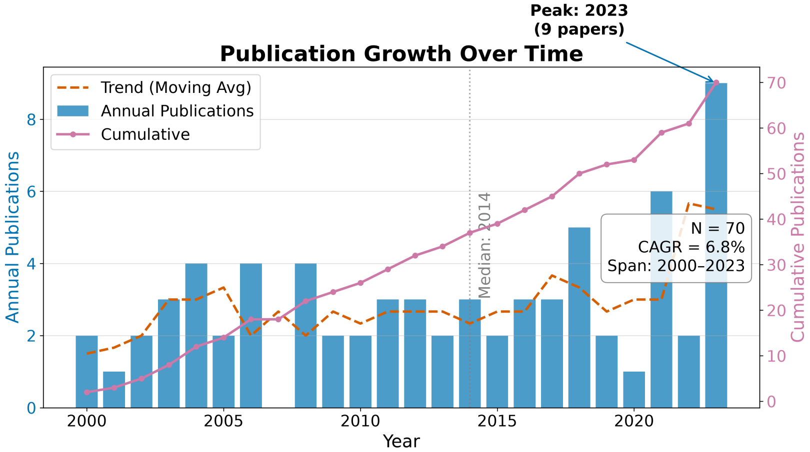


Figure 1: Publication growth curve for Exoplanet Atmospheres. Annual publication counts (bars) and cumulative total (line) show sustained growth from 2000 through 2023, peaking in 2023.

RQ1: Field Size and Growth I

The temporal analysis describes a retained literature slice spanning 23 years. The compound annual growth rate of 6.76% implies a corpus doubling time of approximately 2.1 years under the configured retrieval and indexing conditions. The peak year 2023 contains 9 retained records; that count may reflect both publication activity and delays or differences in source indexing, so it should not be interpreted as a causal trend without a live coverage audit.

RQ1: Field Size and Growth I

Table 1. Top publication years.

RQ1: Field Size and Growth I

Year	Publications
2003	3
2004	4
2006	4
2008	4
2011	3
2012	3
2014	3
2018	5
2021	6
2023	9

RQ2: Subfield Composition I

Records distribute across the 4 configured subfields as shown in Table 2, with **Observational Methods** the largest bucket at 51.4% of the classified corpus. The dominance of Observational Methods is a property of the configured taxonomy and retained corpus. It is not a domain prevalence estimate without a live, source-backed review and a documented coverage assessment.

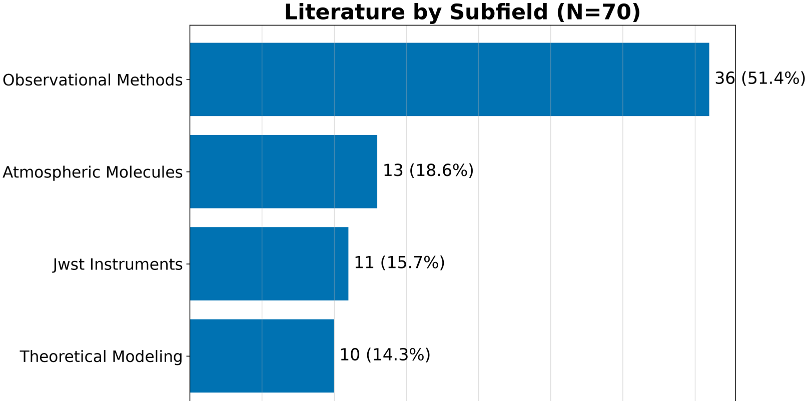
Table 2. Subfield distribution.

RQ2: Subfield Composition I

Subfield	Papers	Share
Observational Methods	36	51.4%
Atmospheric Molecules	13	18.6%
Jwst Instruments	11	15.7%
Theoretical Modeling	10	14.3%

RQ2: Subfield Composition II

Subfield	Papers	Share
Observational Methods	36	51.4%
Atmospheric Molecules	13	18.6%
Jwst Instruments	11	15.7%
Theoretical Modeling	10	14.3%



RQ2: Subfield Composition I

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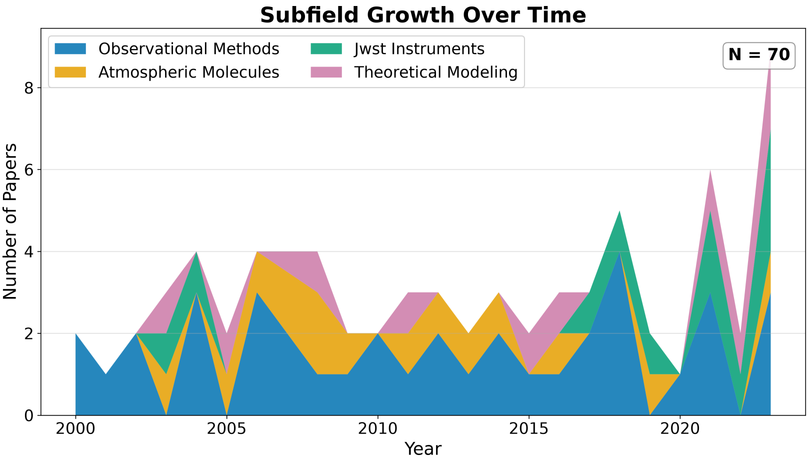


Figure 4: Subfield timeline for Exoplanet Atmospheres. Stacked annual publication counts by subfield show how each sub-area has evolved over time, revealing emerging and declining research threads.

Identifier and Full-Text Coverage I

The corpus has measurable identifier coverage: 80 of 46 records (100.0%) carry DOIs, supporting cross-engine de-duplication. OpenAlex IDs are present for 12 records. Abstract coverage stands at 100.0% (80 records), which limits the text analytics to that subset. Open-access status is confirmed for 42.5% of records, and 42.5% have a direct PDF link.

The corpus spans 144 unique authors across 46 papers, yielding a mean of 1.51 papers per author. Citation counts range from zero to 110 (mean 37.0, median 31.0), with a total of 2,593 citations across the corpus. The Gini coefficient of citation concentration is 0.421. This statistic describes concentration within the retained corpus and should not be generalized to citation behavior in the field.

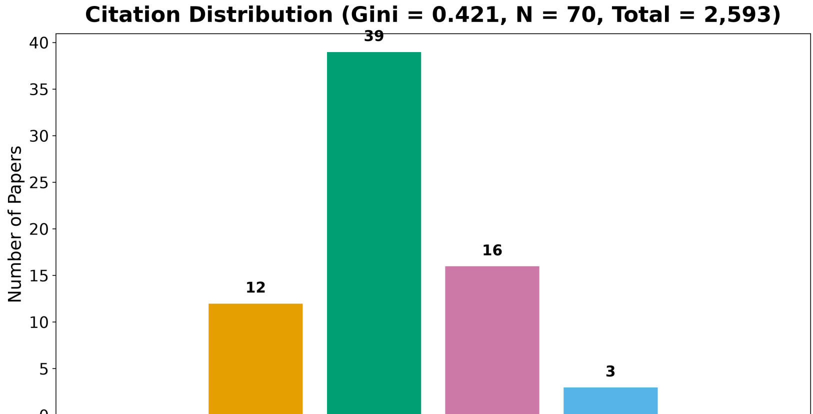
Table 3. Citation count distribution.

Descriptive Bibliometrics I

Citations	Papers
0	0
1-9	12
10-49	39
50-99	16
100-499	3
500+	0

Descriptive Bibliometrics II

Citations	Papers
0	0
1-9	12
10-49	39
50-99	16
100-499	3
500+	0



Descriptive Bibliometrics I

Table 4. Top publication venues.

Descriptive Bibliometrics I

Venue	Papers
Astronomy and Astrophysics	11
Publications of the Astronomical Society of the Pa	11
The Astrophysical Journal	10
Monthly Notices of the Royal Astronomical Society	9
Nature Astronomy	9
Research Notes of the AAS	9
Icarus	6
The Astronomical Journal	5

Descriptive Bibliometrics II

Venue	Papers
Astronomy and Astrophysics	11
Publications of the Astronomical Society of the Pa	11
The Astrophysical Journal	10
Monthly Notices of the Royal Astronomical Society	9
Nature Astronomy	9
Research Notes of the AAS	9
Icarus	6
The Astronomical Journal	5

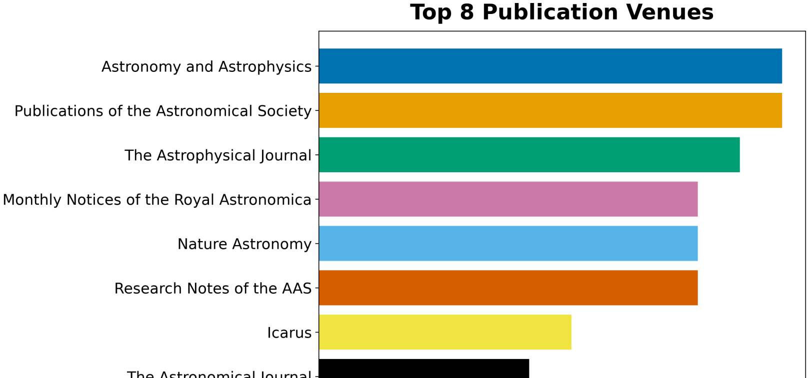


Table 5. Top authors by publication count.

Descriptive Bibliometrics I

Rank	Author	Papers
1	A. Fournier	4
2	B. Owens	4
3	D. Ito	4
4	D. Nakamura	4
5	G. Ito	4
6	K. Ito	4
7	B. Kowalski	3
8	C. Singh	3
9	E. Tanaka	3
10	F. Lindgren	3

Descriptive Bibliometrics II

Rank	Author	Papers
1	A. Fournier	4
2	B. Owens	4
3	D. Ito	4
4	D. Nakamura	4
5	G. Ito	4
6	K. Ito	4
7	B. Kowalski	3
8	C. Singh	3
9	E. Tanaka	3
10	F. Lindgren	3